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$f(x, y) = e^x \sin y$ in $a\left(0, \frac{\pi}{4}\right)$ in de richting $h(1, -1)$

$$f\left(0, \frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$$

$$Df(a, h) = \lim_{\lambda \rightarrow 0} \frac{f\left(\lambda, \frac{\pi}{4} - \lambda\right) - f\left(0, \frac{\pi}{4}\right)}{\lambda}$$

$$= \lim_{\lambda \rightarrow 0} \frac{e^\lambda \sin\left(\frac{\pi}{4} - \lambda\right) - \frac{\sqrt{2}}{2}}{\lambda}$$

Met l'hôpital:

$$= \lim_{\lambda \rightarrow 0} \left(e^\lambda \sin\left(\frac{\pi}{4} - \lambda\right) - e^\lambda \cos\left(\frac{\pi}{4} - \lambda\right) \right)$$

$$= \sin \frac{\pi}{4} - \cos \frac{\pi}{4} = 0$$

References